Eliminative Information in a Generalized Monty Hall Problem II

Variable Reveal Costs, Interior Optima, and Learning Curves

Abstract

Part I studied the generalized Monty Hall problem with a constant cost per host reveal and found an exact endpoint threshold: reveal everything if $c < 1/K$ and reveal nothing if $c > 1/K$. This second note keeps the same eliminative-information mechanism but allows reveal costs to vary with the number of reveals already purchased. The probability of success after t reveals remains

$$S(t) = \frac{m(K-1)}{K(K-1-t)},$$

but the objective becomes $S(t) - C(t)$ for a cumulative cost curve C. The stopping rule is governed by the comparison

$$S(t+1) - S(t) \quad \text{versus} \quad c_t.$$

Linearly increasing costs can create genuine interior optima, so the clean endpoint theorem is not robust to arbitrary learning-cost curves. Costs that rise and then saturate partly restore the Part I "push-through" phenomenon: late eliminative information can again dominate early unattractive steps.

1 The Baseline Carried Forward

There are K doors and m prizes, with $1 \le m < K$. The player chooses one door, and a knowledgeable host can reveal losing doors, never opening a prize or the player's initial door. After t reveals the player switches uniformly among the remaining non-initial closed doors.

The maximum number of possible reveals is

$$T = K - m - 1.$$

The switch success probability is

$$S(t) = \frac{m(K-1)}{K(K-1-t)}, \qquad 0 \le t \le T.$$

Its one-step gain is

$$\Delta_t = S(t+1) - S(t) = \frac{m(K-1)}{K(K-2-t)(K-1-t)}, \qquad 0 \le t < T.$$

The important inherited fact is that Δ_t is strictly increasing. Eliminating a losing door is more valuable when the candidate set is already small.

2 Variable Reveal Costs

Let $c_t \geq 0$ be the cost of buying reveal $t+1$, after t reveals have already been purchased. Define

$$C(t) = \sum_{i=0}^{t-1} c_i, \qquad C(0) = 0.$$

The variable-cost stopping problem is

$$\max_{0 \leq t \leq T} W(t), \qquad W(t) = S(t) - C(t).$$

Proposition 1 (Increment rule)**.** *The net increment from buying one more reveal is*

$$W(t+1) - W(t) = \Delta_t - c_t.$$

Proof. This follows directly:

$$W(t+1) - W(t) = S(t+1) - S(t) - \{C(t+1) - C(t)\} = \Delta_t - c_t.$$

□

Thus variable costs create a competition between two sequences: the increasing benefit sequence Δ_t and the cost sequence c_t.

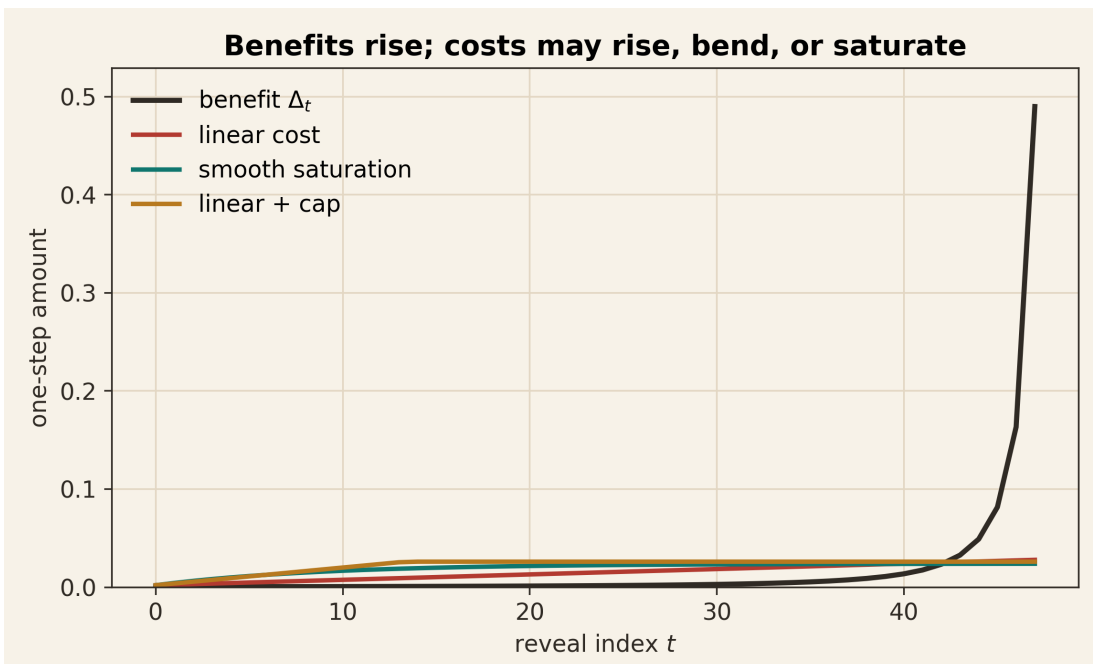

Figure 1: For $K = 50, m = 1$, the eliminative-information benefit Δ_t rises sharply near the end. Different learning-cost curves interact with this same benefit curve in different ways.

3 What Constant Costs Hid

In Part I, $c_t = c$ for every t. Since Δ_t is increasing, the increments $\Delta_t - c$ are also increasing. Therefore W is a discrete convex sequence and its maximum occurs at an endpoint.

This endpoint theorem is a property of constant costs, not of eliminative information alone. Once c_t varies, the increments $\Delta_t - c_t$ need not be monotone.

Theorem 1 (Interior optima are possible)**.** *There exist nonnegative variable costs for which the maximizer of $W(t) = S(t) - C(t)$ is strictly interior.*

Proof. Choose any interior $q \in \{1, \ldots, T-1\}$. Let costs be very small for the first q reveals and very large thereafter. Then W rises up to q and falls after q, so q is an interior maximizer. A linearly increasing cost curve can realize the same qualitative behavior for suitable intercept and slope; Figure 2 gives a concrete example. □

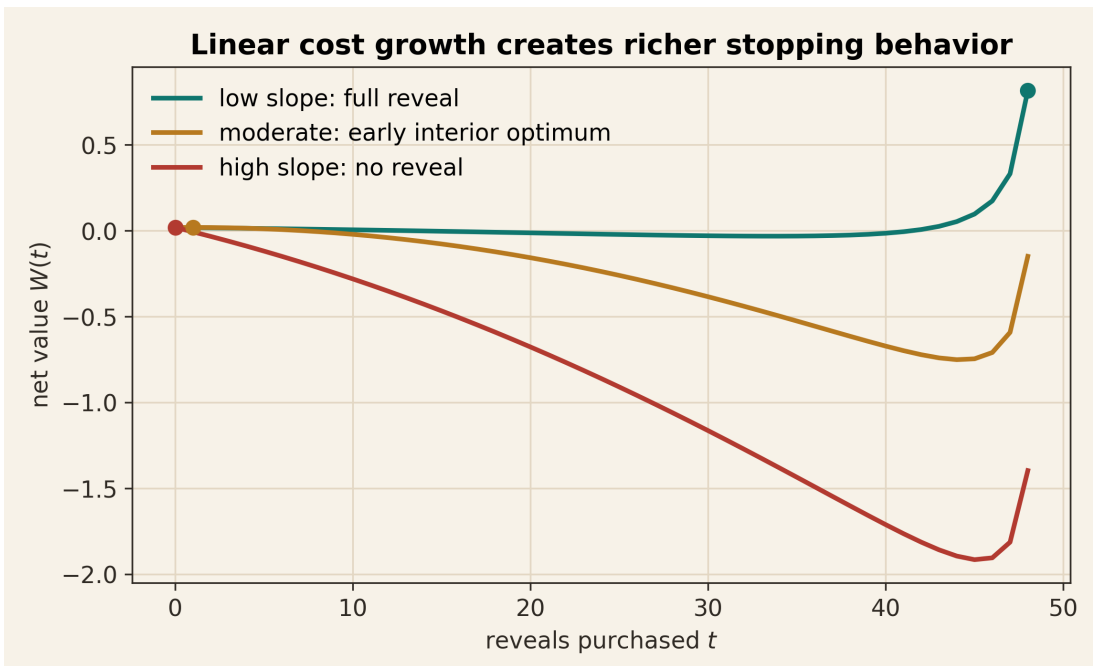

Figure 2: With linearly increasing reveal costs $c_t = a + bt$, the optimum can occur at no reveal, at full reveal, or at an interior stopping time.

4 Linearly Increasing Costs

Consider

$$c_t = a + bt, \qquad a, b \geq 0.$$

Then

$$W(t+1) - W(t) = \frac{m(K-1)}{K(K-2-t)(K-1-t)} - (a + bt).$$

Both terms on the right can increase. The benefit term rises slowly at first and sharply near full revelation; the cost term rises steadily. Their difference can cross zero more than once, creating interior local maxima.

This gives a simple interpretation:

- If a and b are low, late information dominates and full reveal wins.
- If a is high, buying even the first reveal is unattractive.
- If early costs are low but b is high enough, the first few reveals can be worthwhile while continued revealing becomes too expensive, producing an interior optimum.

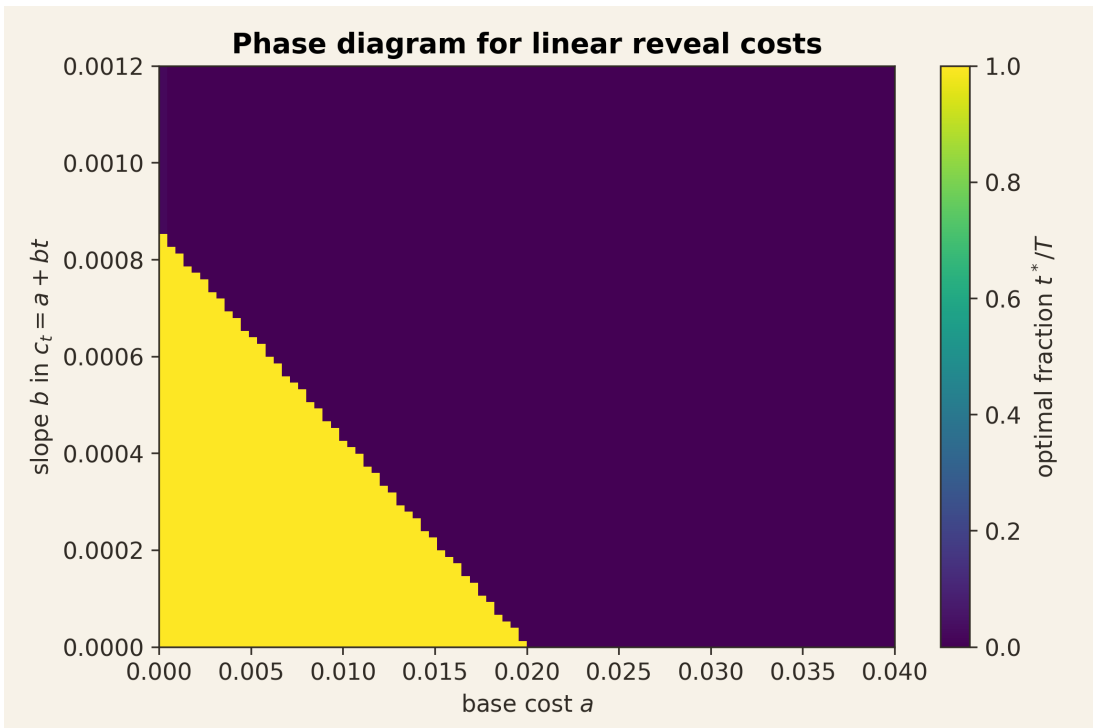

Figure 3: A phase diagram for $K = 50, m = 1$ and $c_t = a + bt$. Color indicates the optimal fraction of possible reveals purchased.

5 Costs That Rise and Then Saturate

In many learning processes, costs rise early but do not rise forever. There may be setup costs, attention costs, or search-friction costs that plateau once the process is underway. Two simple forms are

$$c_t = a + \min\{bt, L\}$$

and

$$c_t = a + L(1 - e^{-t/\tau}).$$

These costs initially increase, but eventually flatten. Because Δ_t continues to rise sharply near the end, saturated costs can restore the push-through effect: after paying through the early region, late reveals become increasingly attractive again.

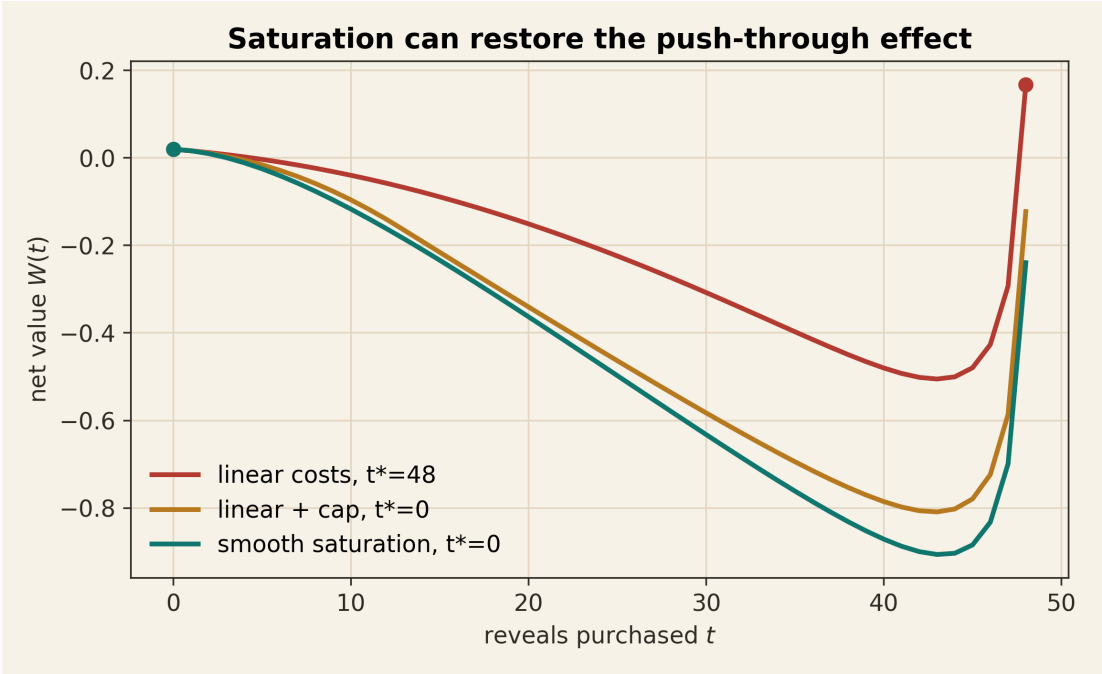

Figure 4: Compared with purely linear costs, saturating cost curves can make late-stage full revelation optimal again.

6 Interpretation

Variable costs separate two ideas that coincide in the constant-cost model. The first idea is robust: eliminative information has increasing marginal value. The second idea is fragile: endpoint optimality. Endpoint optimality requires the net increments to have the right monotonic shape. Constant costs guarantee that shape; arbitrary variable costs do not.

The useful organizing principle is therefore not simply "reveal everything or reveal nothing." It is:

$$\text{continue when } \Delta_t \text{ is large relative to } c_t.$$

Since Δ_t is back-loaded, the value of a reveal can be low early and high late. Whether the player reaches the high-value region depends on the cost curve.

7 Conclusion

The exact $1/K$ threshold from Part I is a clean benchmark, but it is not the end of the story. Variable reveal costs generate a richer stopping geometry. Linearly increasing costs can produce interior optima. Costs that rise and saturate can restore the Monty-style push-through effect. Across these variants, the central mechanism remains the same: eliminative information concentrates probability mass, so its marginal value rises as the set of remaining possibilities shrinks.

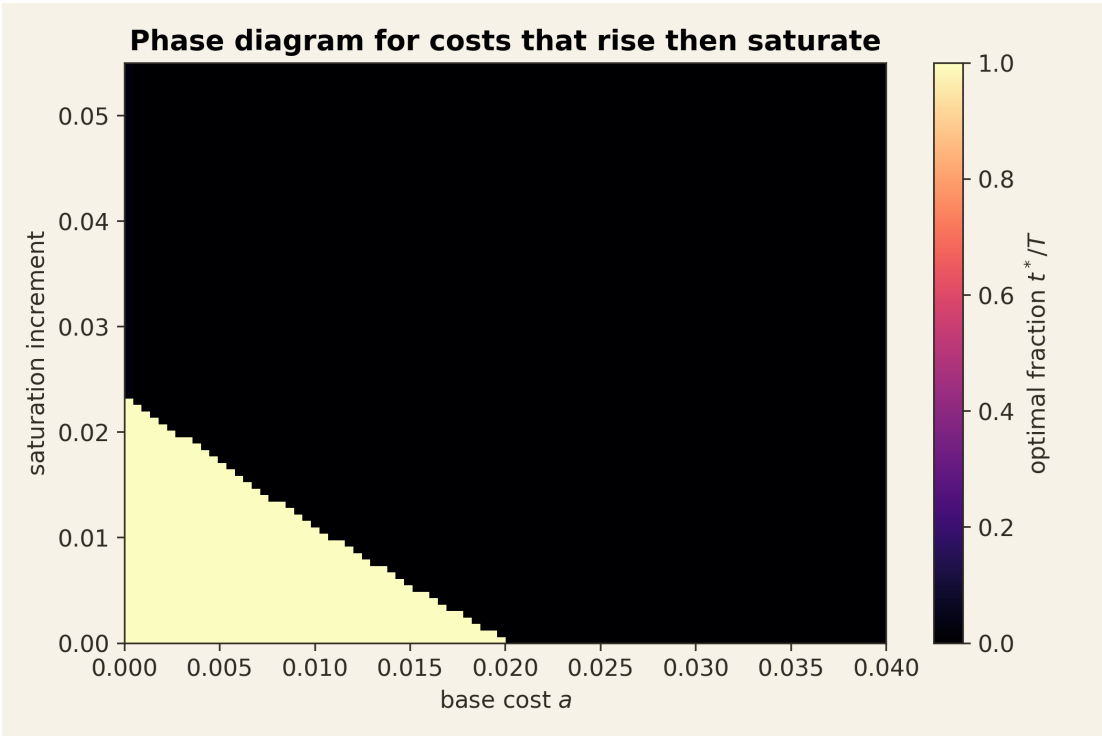

Figure 5: A phase diagram for costs of the form $c_t = a + \min\{bt, L\}$, with fixed slope and varying base cost and saturation level.